

The Quasi-Periodic Diffraction Spectrum of Twin-Pair Correlation on the $6N$ Skeleton

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Abstract

Parts XX–XXII derived, in closed form, the spatial correlation of twin centres on the $6N$ skeleton as an infinite product $R(j) = \prod_{q>3} R_q(j)$ of quantized single-frequency interference factors. This paper renders that closed form directly: evaluating the three interference states across the spatial-frequency spectrum produces a two-panel *diffraction spectrum* of the lattice—a heat map of the single-frequency tensor $R_q(j)$ and the cumulative correlation $R(j)$. The picture makes three established analytic facts legible at a glance: the $O(1/q^2)$ decay of high-frequency amplitude (so low primes dominate the structure); the maximal two-body *suppression* at $j = 1$, where the cumulative product falls to $R(1) = \mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2 \approx 0.41$ (strictly positive—a suppressed but admissible quadruplet, *not* annihilation); and the multi-frequency *super-resonance* at $j = 35 = 5 \cdot 7$, where State A fires at both $q = 5$ and $q = 7$ and $R(35) \approx 2.26$. The rendering is an illustration of the analytic $R(j)$, equal at integer lags to the discrete singular-series ratio; it is not new prime data.

1 Introduction

In Parts XX through XXII, we established that the spatial correlation between twin centres at distance j is not random but is governed by a circular convolution of castellated indicators on the phase-space tori, with the global correlation given by the absolutely convergent product $R(j) = \prod_{q>3} R_q(j)$ and a finite spectral cutoff $k^*(\varepsilon)$ [1, 2, 3].

In this paper we render that closed-form correlation visually. Plotting the tensor of single-frequency interference states and their cumulative product yields the “diffraction spectrum” of the lattice—a compact way to read off the small-prime dominance, the suppression and resonance lags, and the quasi-periodic structure that the analytic results predict.

2 The Tensor Construction

We construct the single-frequency interference tensor $\mathbf{R} = [R_q(j)]$ for twin–twin correlation. Following the exact convolution states of Part XX [1], each entry takes one of three values according to the shift j and frequency q :

1. **State A (Constructive Resonance):** $R_q(j) = \frac{q}{q-2}$ when $j \equiv 0 \pmod{q}$.
2. **State B (Partial Interference):** $R_q(j) = \frac{q(q-3)}{(q-2)^2}$ when $j \equiv \pm 3^{-1} \pmod{q}$.
3. **State C (Destructive Misalignment):** $R_q(j) = \frac{q(q-4)}{(q-2)^2}$ for all other j .

3 Holographic Rendering and Phenomenological Verification

We render this tensor as a two-panel interferogram (Figure 1), showing both the single-frequency spectral map and the cumulative correlation $R(j)$. The picture illustrates—it does not independently prove—three features established analytically in earlier parts:

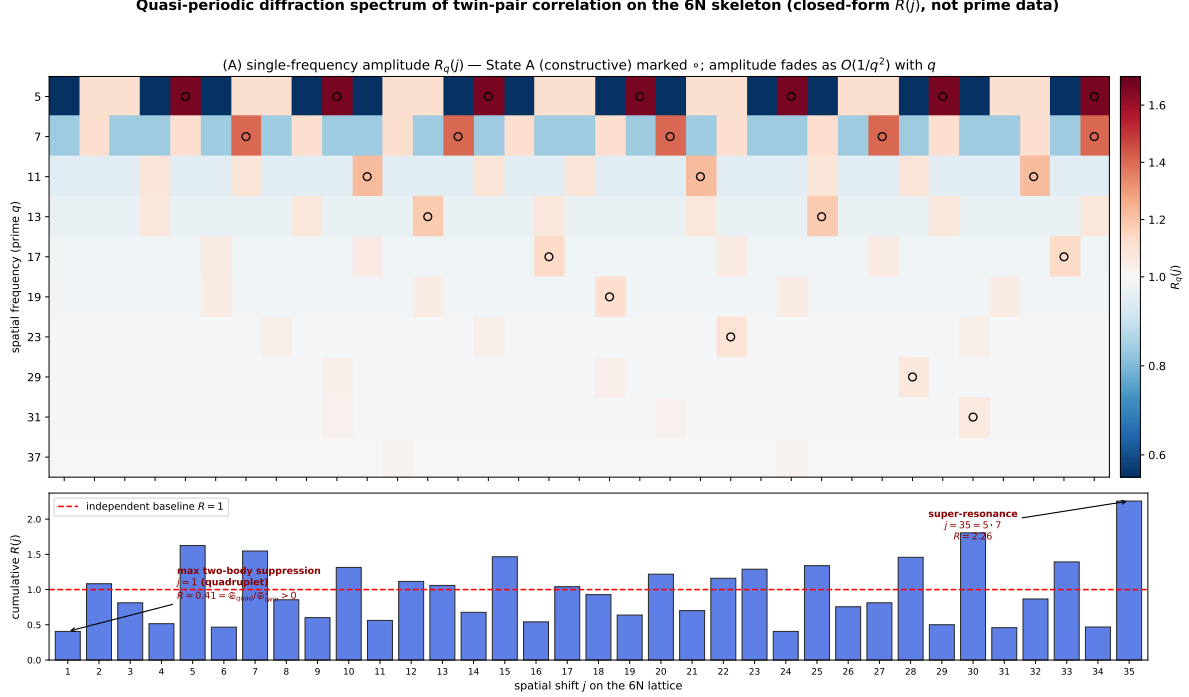


Figure 1: The diffraction spectrum of twin-pair correlation (closed-form $R(j)$, *not* prime data). (A) Heat map of the single-frequency amplitudes $R_q(j)$; State A (constructive, $q \mid j$) cells are circled, and amplitude fades as $O(1/q^2)$ down the frequency axis. (B) The cumulative correlation $R(j) = \prod_q R_q(j)$: maximal two-body suppression at $j = 1$ ($R = \mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2 \approx 0.41 > 0$) and super-resonance at $j = 35 = 5 \cdot 7$ ($R \approx 2.26$), about a baseline of 1.

3.1 Spectral decay

Reading the heat map (Figure 1A) down the frequency axis (increasing q), the colour fades rapidly toward neutrality ($R_q \approx 1$). This illustrates the absolute-convergence result of Part XXII [3]: away from its finitely many resonances each frequency sits in State C with deviation $4/(q-2)^2 = O(1/q^2)$, so the macroscopic structure of $R(j)$ is dominated by the lowest frequencies ($q \leq 13$) while the high-frequency tail contributes a small, summable correction.

3.2 Maximal suppression at $j = 1$

At $j = 1$ —two twin centres one step apart, i.e. a prime quadruplet—no prime is in a constructive state: State A needs $q \mid 1$ and State B needs $q \mid (3 \mp 1)$, both impossible for $q > 3$, so *every* frequency sits in State C. The cumulative product therefore reaches its minimum, $R(1) \approx 0.41$, below even the single-frequency floor $R_5(1) = 5/9 \approx 0.556$.

This is the point most easily misread, so we state it carefully. $R(1) \approx 0.41$ is a *suppression*, not an annihilation: it is strictly positive, and equals exactly the quadruplet prefactor $R(1) = \mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2$, the intercept-suppression identified in Part XX [1]. The quadruplet is an admissible constellation with positive density; the dip merely says two adjacent twins are *anti-correlated* relative to independence. Genuine zero-measure annihilation ($R = 0$) requires

the dead-residue sets to *tile* $\mathbb{Z}/q$, which a twin pair can never achieve—the twin is immune to two-body annihilation, and the lowest order at which annihilation occurs is three-body (Part XXI) [2]. The $j = 1$ trough is thus the deepest *constructive-free* suppression, not a collapse to zero.

3.3 Multi-frequency super-resonance at $j = 35$

At $j = 35 = 5 \cdot 7$, State A (constructive, $R_q = q/(q - 2)$) fires simultaneously at $q = 5$ and $q = 7$, since both divide 35. Their factors $\frac{5}{3}$ and $\frac{7}{5}$ multiply against an otherwise near-neutral tail, lifting the cumulative product to its maximum in the window, $R(35) \approx 2.26$. This is the constructive mirror of the $j = 1$ trough: a lag whose small-prime structure aligns the dead-residue sets of the two centres rather than scattering them. Such resonances recur whenever j is divisible by several small primes (the quasi-periodicity of $R(j)$).

4 Conclusion

Rendering the single-frequency tensor $R_q(j)$ and its cumulative product $R(j)$ makes the closed-form results of Parts XX–XXII legible in one picture: a quasi-periodic, small-prime-dominated correlation that behaves like a multi-frequency diffraction pattern, with anti-correlated suppression troughs (the deepest at $j = 1$, the quadruplet prefactor $\mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2 \approx 0.41 > 0$) and constructive resonances (the strongest here at $j = 35 = 5 \cdot 7$, $R \approx 2.26$). The rendering is an illustration of the analytic correlation, equal at integer lags to the discrete singular-series ratio; it adds no new prime data and makes no claim about the infinitude of any constellation.

Data and code availability. The figure-generating script is openly available at <https://github.com/Ruqing1963/6N-diffraction-spectrum> [5].

References

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